<u>**PROJECT 07 REPORT**</u>

Ultrasonic Distance Measurement System using HC-SR04 Sensor

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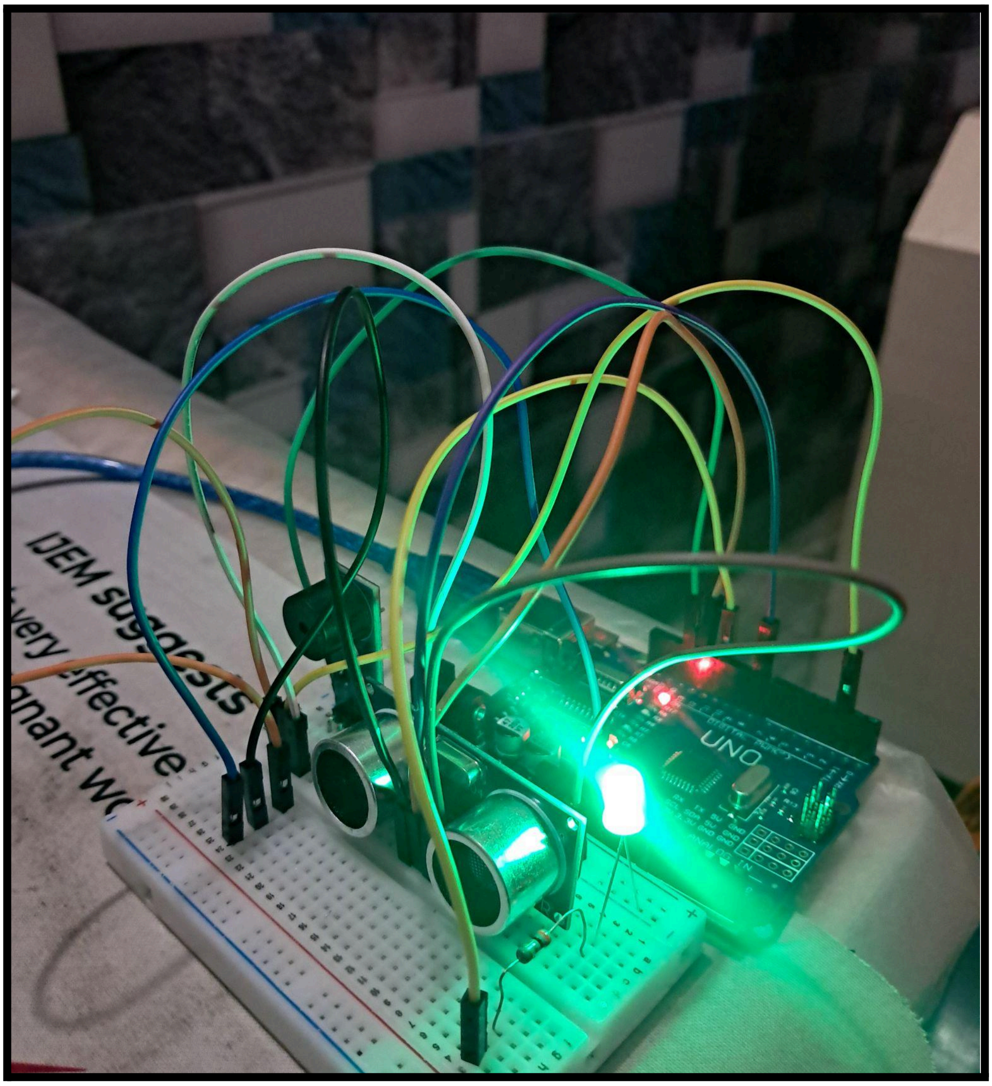
Duration : 2 days



1. Objective

The objective of this project was to develop a distance measurement system using HC-SR04 ultrasonic sensor and Arduino Uno.

The project also introduced ultrasonic sensing, Time of Flight, pulse duration measurement, speed of sound calculations, trigger echo communication and distance based decision making.

2. Components Used

Component	Quantity
Arduino Uno R3	1
Breadboard	1
HC-SR04 Ultrasonic Sensor	1
Active Buzzer Module	1
LED (Green)	1
Resistors (300Ω)	1
Jumper Wires	Multiple
USB Cable	1

3. Software Used

- Arduino IDE
- Serial Monitor
- Windows Laptop

4. Theory

The HC-SR04 is an ultrasonic distance sensor capable of measuring the distance of nearby objects using ultrasonic sound waves operating at approximately 40 kHz.

The sensor consists of two ultrasonic transducers:

- **Transmitter (T)**
    - Generates ultrasonic sound pulses
- **Receiver (R)**
    - Receives the reflected ultrasonic waves after they bounces back from an object

The HC-SR04 contains four pins:

1. VCC
2. TRIG
3. ECHO
4. GND

Arduino measures this HIGH pulse duration using:
pulseIn(ECHO pin, HIGH);

The measured time is converted into distance using:

$$\textbf{Distance} = (\textbf{Duration} \times 0.0343) / 2$$

where,

- ❖ 0.0343 cm/µs is the speed of sound converted into centimeters per microsecond.
- ❖ Division by 2 accounts for the round trip travel of the ultrasonic wave.

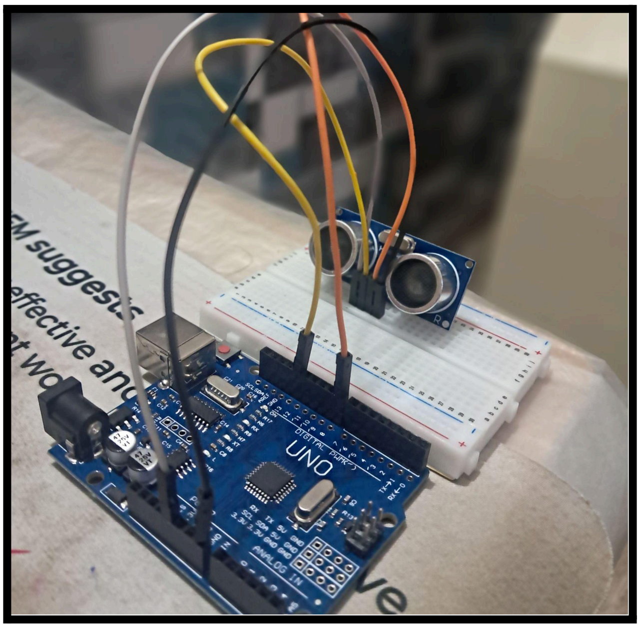
5. Tasks Performed

➢ Understanding the HC-SR04 Ultrasonic Sensor

The HC-SR04 ultrasonic sensor was studied to understand its operating principle, pin configuration, transmitter and receiver transducers and ultrasonic distance measurement technique.

The concept of Time of Flight (TOF) was explored to understand how ultrasonic waves are transmitted, reflected, and converted into measurable distance.

➢ Distance Measurement using Serial Monitor



Video

The HC-SR04 sensor was interfaced with Arduino Uno to continuously measure object distance.

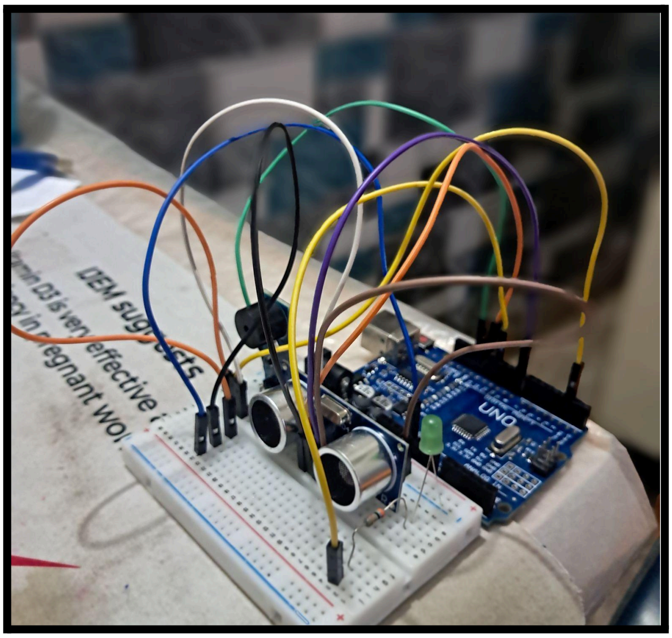
Observed behavior:

Flat Cardboard Surface → Accurate and Stable Measurements

Soft Toy → Fluctuating Readings due to weak ultrasonic reflections

The measured distance was continuously displayed on the Serial Monitor.

➢ Distance Based Object Detection Alarm



Video

A proximity alarm system was implemented using the measured distance from the HC-SR04 sensor.

Whenever the object crossed threshold distance (9cm), the LED and Buzzer turned ON and as the object distanced greater than the threshold distance the LED and Buzzer automatically turned OFF.

6. Learning Outcomes

During this project, the following concepts were learned:

- HC-SR04 Ultrasonic Sensor Operation
- Ultrasonic Sound Waves
- Time of Flight (TOF)
- Trigger and Echo Communication
- pulseIn() Function

- Speed of Sound Conversion
- Distance Calculation
- Microsecond Timing
- Threshold Based Object Detection
- Distance Based Alarm Systems
- Effect of Object Material on Ultrasonic Reflection

7. Problems Faced

I. Distance readings were inaccurate while testing with a soft toy. **(A flat cardboard surface was used for testing as soft and irregular surfaces absorb and scatter ultrasonic waves)**

8. Observations

- The HC-SR04 accurately measured the distance of nearby hard and flat objects.
- Ultrasonic waves reflected more effectively from rigid surfaces than soft materials.
- Soft objects produced weaker echoes which resulted in fluctuation in readings.
- The measured distance corresponds to the nearest object within the sensor's detection cone.
- The ECHO pulse duration increased as the object moved farther away.
- Accurate distance measurement depends on proper ultrasonic wave reflection.

- The distance based alarm responded reliably within the selected threshold distance.

9. Applications and Future Scope

Ultrasonic distance sensors are widely used in robotics, industrial automation, obstacle detection, parking assistance systems, water level monitoring, smart dustbins, security systems and autonomous navigation.

Possible future upgrade includes Servo based 180° Radar Scanner,OLED Distance Display,Multi Sensor Obstacle Detection,Automatic Parking Assistance System and Autonomous Robot Navigation.

10. Conclusion

This project successfully demonstrated the development of an ultrasonic distance measurement and object detection system using the HC-SR04 sensor and Arduino Uno.

The project represents the first **distance measurement and ranging system** in the portfolio and provides a strong foundation for the future.